Mirza Mehmedagic

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EDUCATION

UNIVERSITY OF CHICAGO Chicago, IL
Attended: Sept 2022 - June 2026 GPA: 3.97/4.00
Programs: BS in Computer Science (specialization in theory), BS in Mathematics, MS in Computer Science
Coursework: Systems Programming, Bioinformatics, Complexity Theory, Information and Coding Theory, Discrete Mathematics, Linear Algebra, Real Analysis, Point-Set Topology, Abstract Algebra, Mathematical Logic, Markov Chains and Stochastic Processes

LANE TECH COLLEGE PREP Chicago, IL
Attended: Sept 2018 - June 2022 GPA: 4.00/4.00
Coursework: AP Computer Science A, AP Calculus BC, Multivariable Calculus, AP Statistics, AP Physics C

PROFESSIONAL EXPERIENCE

Department of Computer Science, University of Chicago Chicago, IL
Graduate Student Teaching Assistant Sep - Dec 2025, Mar - Jun 2026
- Hosted weekly office hours and midterm exam review session that often consisted of demonstrating proofs from undergraduate complexity theory and formal languages courses and interactive homework Q&A with multiple students at once
- Graded proofs-based homework exercises/exam questions and answered online forum questions in a timely manner

Biological Sciences Division, University of Chicago Chicago, IL
Undergraduate Student Teaching Assistant Sep - Dec 2023, Jan - Mar 2024, Sep - Dec 2024, Mar - Jun 2025
- Produced R-based lab assignments for the sequence of classes titled Multiscale Modelling of Biological Systems (genomics, DNA analysis, proteomics, molecular dynamics, etc.) and a mathematical modeling for pre-med students
- Graded these labs for feedback for students and hosted open lab sessions/office hours for these labs

Gelber Group LLC Chicago, IL
Trading Intern May - Jul 2023
- Used data science and quantitative algorithmic principles to develop a trading strategy using financial market data
- Presented findings on volatility spillover from foreign markets into the US open and close to a team of algorithmic traders

Hyde Park Math and Reading Center, Kumon Chicago, IL
Learning Assistant Jul 2022 - Jun 2023
- Graded students' math and reading worksheets and conducted primary instruction of math and reading concepts to students
- Aided supervisor in managing students during primary instruction

RESEARCH EXPERIENCE

"The Zig-Zag Pseudorandom Generator", University of Chicago Computer Science MS Thesis. Sep 2025 - Jun 2026
- Master's thesis project focused on studying the pseudorandom walk generator resulting from Reingold's log-space algorithm for deciding undirected graph connectivity and what computational models (e.g., permutation branching programs) it can fool
- Presented several proofs from readings to thesis advisor William Hoza, including one that resolved a potential research question on the gap between certificate complexity and average decision tree depth of a Boolean function

"Markov Chain Mixing and Random Walks on Regular Graphs", University of Chicago Math REU Jun - Sep 2025
- REU project focused on the theory of Markov chain mixing, focusing on both coupling and spectral techniques and culminating in a study of a 2010 paper on mixing of random walks on regular graphs (mentored by Stephen Yearwood)
- Culminated in an expository research paper: https://math.uchicago.edu/~may/REU2025/REUPapers/Mehmedagic.pdf (may be under revision and thus a dead link)
- Attended lectures concerning orthogonal polynomial theory (with applications to computer science), Lebesgue differentiation theory, ergodic theory, and more

"Geometric Measure Theory with Methods from Algorithmic Complexity", University of Chicago Math REU Jun - Sep 2024
- REU project focused on various characterizations of fractal dimension including classical definitions and the point-to-set principle (mentored by Donald Stull)
- Culminated in an expository research paper: https://math.uchicago.edu/~may/REU2024/REUPapers/Mehmedagic.pdf (may be under revision and thus a dead link)
- Attended lectures concerning stochastic processes and random walks, representation theory of finite groups, a survey of results for random trees, and more

VOLUNTEER EXPERIENCE

University of Chicago Science Olympiad Chicago, IL

Member Oct 2022 - Present

- Supervised at events for and wrote science tests for 4 invitational competitions hosted by the University of Chicago as well as for 4 regional competitions and 2 state competitions hosted by the parent organization Illinois Science Olympiad
- Focused said supervising and writing on an event centered on the physics and engineering of wind power, involving great
- Developed sufficient expertise to demonstrate the wind power event (including supervision of testing students' self-built devices) at the 2024 Great Lakes Coaches' Clinic

Webmaster

- Maintained the student organization's website using HTML and WordPress Aug 2023 - Aug 2025

CodeLab, Lane Tech College Prep Chicago, IL

Member Sep 2018 - May 2022

- Attended weekly meetings to practice programming skills and to join various teams of like-minded individuals that participated in programming competitions (Procom 2020) and virtual CTF competitions

Co-President Oct 2021 - May 2022

- Organized weekly activities for interested students that served as workshops for learning to code in Python, understanding basic logic design, and using Linux and Git, etc.

TECHNICAL PROJECTS

HackIMSA, Illinois Math and Science Academy May 2022

- 4-member team awarded "Best in Education" for Safari Squabble: https://github.com/TheApplePieGod/IMSAHackathon
- Built an online game for children to develop their logical and analytical thinking skills with React and Typescript

CodeDay, Student Research and Development Sep 2019

- 2-member team awarded "Best in Class" for Drenchspire: https://github.com/hstennes/Drenchspire
- Built an adaption of a color-based logic puzzle game for the TI-nSpire calculator with Lua

HackChicago, Execute Big Jul 2018

- 2-member team awarded runner-up in "Best Game" for Bad Magic: https://github.com/mmehmedagic/BadMagic
- Built a two-person, one-keyboard game (code, graphics, etc.) with Java

DISTINCTIONS

University of Chicago:

- **Dean's List:** 2023, 2024, 2025
- **Robert Maynard Hutchins Scholar:** 2024
- **Enrico Fermi Scholar:** 2025, 2026
- **Summa Cum Laude**

Elected to: Beta of Illinois Chapter of Phi Beta Kappa

SKILLS

Technical:

- **Programming:** Proficient (and university-certified) in Python, proficient in R, proficient in Java, experienced in C/C++
- **Tools:** Git, Vim, Visual Studio Code, Android Studio, IntelliJ, Microsoft Office
- **Frameworks / Libraries:** TensorFlow, ReactJS

Language: Proficient in Bosnian/Croatian/Serbian (reading/writing/speaking), experienced in Spanish (reading/writing/speaking)